BIOST 2155: Assignment 2

YOUR NAME HERE (YOUR PITT EMAIL HERE)

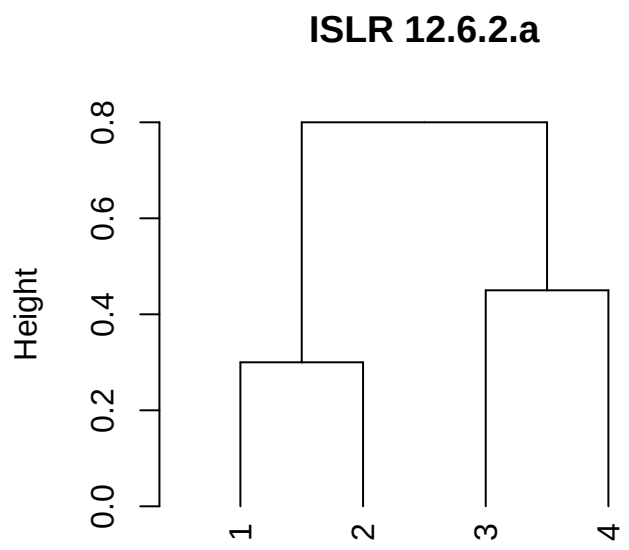
Released: 2025-10-24. **Due by**: 2025-11-09, 11:59 PM

Theory (20 + 10 = 30 points)

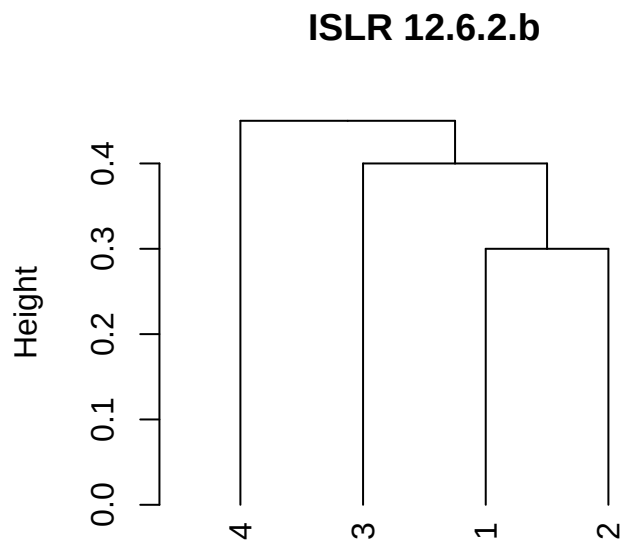
Part A (20 points)

ISLR 12.6.2, page 546. Please attempt all five parts. 7 points each for parts (a) and (b) and 2 points each for parts (c), (d), and (e). Show your working - you may upload handwritten answers for this, I don't expect you to type out dendrograms.

(a)



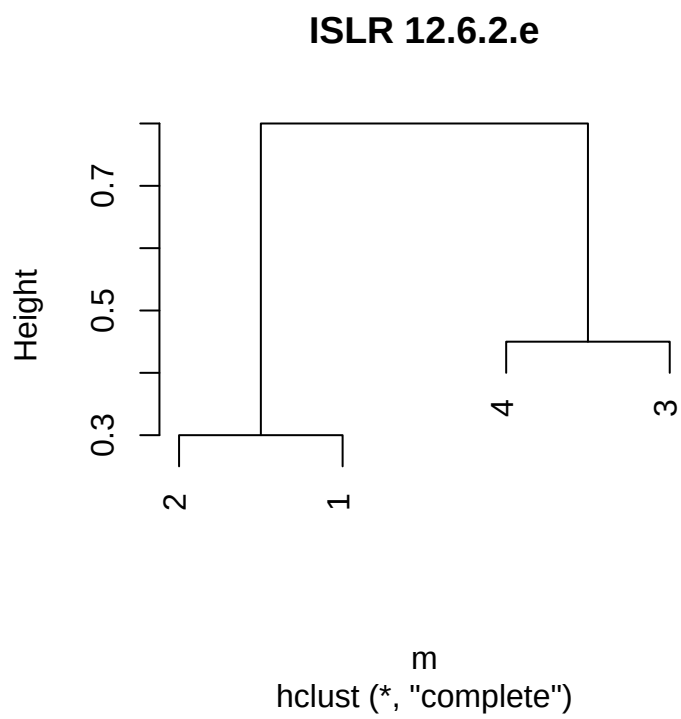
(b)



(c) **(1,2) & (3,4)**

(d) **((1,2),3) & 4**

(e)



Part B (10 points)

This problem involves the K-means clustering algorithm. Please attempt both parts. 7 points for part (a) and three points for part (b). Show your working - you may upload handwritten answers for this as well. Let $C_1, \ldots, C_K$ be a clustering of n data points. The objective function of K-means is given by

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} n(C_k)^{-1} \sum_{j,j' \in C_k} \sum_{i=1}^{p} (x_{ji} - x_{j'i})^2 \right\},$$

where $n(C_k)$ is the number of data points in C_k.

(i) Given that the center of C_k is $\boldsymbol{\mu}_k = n(C_k)^{-1} \sum_{j \in C_k} \boldsymbol{x}_j$, prove that the objective function above is equivalent to

$$\min_{(C_1,\ldots,C_K)} \sum_{k=1}^{K} \left\{ 2 \cdot \sum_{j \in C_k} \sum_{i=1}^{p} (x_{ji} - \mu_{ki})^2 \right\}.$$

```
### answer part (i) starts
```

We have the following equation:

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_{kj})^2$$

Where, $\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the mean for feature j in cluster C_k. Expanding the LHS over i:

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 = \frac{1}{|C_k|} \sum_{i' \in C_k} \sum_{j=1}^{p} (x_{1j} - x_{i'j})^2 + \frac{1}{|C_k|} \sum_{i' \in C_k} \sum_{j=1}^{p} (x_{2j} - x_{i'j})^2 + \ldots$$

Further expanding a single term from the summation:

$$\begin{aligned}
&\frac{1}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \\
&= \frac{1}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} ((x_{ij} - \bar{x}_j) - (x_{i'j} - \bar{x_j}))^2 \\
&= \frac{1}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} ((x_{ij} - \bar{x}_j)^2 - 2(x_{ij} - \bar{x}_j)(x_{i'j} - \bar{x_j}) + (x_{i'j} - \bar{x_j})^2) \\
&= \frac{1}{|C_k|} \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_j)^2 - \frac{2}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} (x_{ij} - \bar{x}_j)(x_{i'j} - \bar{x_j}) + \frac{1}{|C_k|} \sum_{i' \in C_k} \sum_{j=1}^{p} (x_{i'j} - \bar{x_j})^2 \\
&= \frac{2}{|C_k|} \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_j)^2 - \frac{2}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} (x_{ij} - \bar{x}_j)(x_{i'j} - \bar{x_j})
\end{aligned}$$

Substituting the final term from above in the summation:

$$2\frac{|C_k|}{|C_k|} \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_j)^2 \underbrace{- \frac{2}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} (x_{1j} - \bar{x}_j)(x_{i'j} - \bar{x_j}) - \frac{2}{|C_k|} \sum_{i' \in C_k, i' \neq i} \sum_{j=1}^{p} (x_{2j} - \bar{x}_j)(x_{i'j} - \bar{x_j}) + \ldots}_{\text{This term is equal to zero because } \sum_{i' \in C_k} x_{i'j} - |C_k| \bar{x_j} = 0}$$

The remaining term is:

$$2\frac{|C_k|}{|C_k|}\sum_{i\in C_k}\sum_{j=1}^{p}(x_{ij}-\bar{x}_j)^2$$

```
### answer part (i) ends
```

(ii) On the basis of the identity proved in (i), argue that Lloyd's clustering algorithm decreases the above objective function at each iteration.

```
### At end of each iteration the K-Means algorithm assigns the
### observations to the closest (Euclidean distance) centroid.
### Minimising the RHS of the identity will reduce the euclidean
### distance and so the LHS (10.11) will also be reduced. In other
### words the identity shows that minimizing the sum of the squared
### Euclidean distance for each cluster is the same as minimizing the
### within-cluster variance for each cluster.
```

Application (50 + 20 points = 70 points)

You are tasked with analyzing a new dataset on maternal health. The dataset contains physiological measurements (blood pressure, heart rate, etc.) for `n = 1,014` patients and a clinical assessment of their pregnancy risk level (`high risk`, `mid risk`, and `low risk`). Your goal is to explore this data, build predictive models to classify patient risk, and investigate the underlying structure of the data. **Please set a random seed (`set.seed(42)`) before any step involving data splitting or randomization to ensure your results are reproducible.**

Data has been collected from different hospitals, community clinics, maternal health cares from the rural areas of Bangladesh through the IoT based risk monitoring system. Below is a description of the variables contained in the dataset.

- `Age`: Age of the patient in years.
- `SystolicBP`: The upper value of blood pressure, measured in mmHg.
- `DiastolicBP`: The lower value of blood pressure, measured in mmHg.
- `BS`: Blood Sugar levels, measured in mmol/L.
- `BodyTemp`: Body Temperature, measured in Fahrenheit.
- `HeartRate`: Resting heart rate, measured in beats per minute (BPM).
- `RiskLevel`: The predicted clinical risk level during pregnancy (Low, Mid, or High).

```r
rm(list = ls())
pacman::p_load(tidyverse, caret, boot, here, corrplot, tableone, biotools)
data <- read_csv(file.path("assign2_classData.csv"))
```

Question 1: Initial Data Exploration (10 Points)

Before building any models, it's crucial to understand the data. Create a table 1 that summarizes the patient characteristics, stratified by `RiskLevel`. You may use the `tableone` package for this. Please report and interpret mean ± standard deviation for all continuous variables and a column for the overall p-value to test for differences between the three groups. Based on your table 1 p-values, which patient variables show a statistically significant difference across the three `RiskLevel` groups?

```r
### table1 ####
covars <- c("Age", "SystolicBP", "DiastolicBP", "BS", "BodyTemp", "HeartRate")
table1 <- CreateTableOne(vars = covars, strata = "RiskLevel", data = data,
    test = TRUE)
print(table1, test = TRUE, pDigits = 3, noSpaces = TRUE)
```

```
##                          Stratified by RiskLevel
##                           high risk      low risk       mid risk       p
##   n                       272            406            336
##   Age (mean (SD))         36.22 (13.03)  26.87 (13.12)  28.36 (12.55)  <0.001
##   SystolicBP (mean (SD))  124.19 (20.23) 105.87 (15.89) 113.15 (14.98) <0.001
##   DiastolicBP (mean (SD)) 85.07 (14.11)  72.53 (13.05)  74.23 (11.49)  <0.001
##   BS (mean (SD))          12.12 (4.17)   7.22 (0.65)    7.80 (2.29)    <0.001
##   BodyTemp (mean (SD))    98.90 (1.56)   98.37 (1.11)   98.83 (1.43)   <0.001
##   HeartRate (mean (SD))   76.74 (8.70)   72.77 (8.29)   74.18 (6.77)   <0.001
##                          Stratified by RiskLevel
##                           test
##   n
##   Age (mean (SD))
##   SystolicBP (mean (SD))
##   DiastolicBP (mean (SD))
##   BS (mean (SD))
##   BodyTemp (mean (SD))
##   HeartRate (mean (SD))
```

Question 2: Checking Statistical Assumptions (10 Points)

We want to build a classifier like Linear or Quadratic Discriminant Analysis (LDA/QDA). These models have statistical assumptions about the data that we must check first.

Part A (3 points) Perform a one-way `MANOVA` to test the null hypothesis that the mean vectors (the combined average of all 6 predictors) are the same across all three `RiskLevel` groups. Based on the p-value, do we reject or fail to reject the null hypothesis? What does this result tell us about the patient groups?

```r
### equality of means ####
manova_result <- manova(as.matrix(data |>
    dplyr::select(-"RiskLevel")) ~ data |>
    pull("RiskLevel"))
summary(manova_result)
```

```
##                           Df  Pillai approx F num Df den Df    Pr(>F)
## pull(data, "RiskLevel")    2 0.54479   62.833     12   2014 < 2.2e-16 ***
## Residuals               1011
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the p-value, which is less than 0.05, we REJECT the null hypothesis. This result tells us that there is a statistically significant difference in the multivariate sample means of the 6 predictors across the three `RiskLevel` groups.

Part B (5 points) Use `Box's M` test to check the assumption of homogeneity of dispersion matrices (i.e., if the variance-covariance matrices are the same) across the three groups. Based on the p-value, are the dispersion matrices the same or different?

```r
### equality of dispersion matrices ####
box_m_test_result <- boxM(as.matrix(data |>
    dplyr::select(-"RiskLevel")), data |>
    pull("RiskLevel"))
print(box_m_test_result)
```

```
##
##  Box's M-test for Homogeneity of Covariance Matrices
##
## data:  as.matrix(dplyr::select(data, -"RiskLevel"))
## Chi-Sq (approx.) = 1238.3, df = 42, p-value < 2.2e-16
```

Based on the p-value, which is less than 0.05, we REJECT the null hypothesis. This result tells us that the assumption of homogeneity of covariance matrices is VIOLATED. In other words, this means the dispersion matrices (the variance/covariance) of the 6 predictors are statistically different across the three `RiskLevel` groups.

Part C (2 points) Given your findings from (A) and (B) combined, which model is the more appropriate choice: LDA or QDA? Briefly explain why.

In part (A), the significant MANOVA result confirmed that the mean vectors (the combined averages of the 6 predictors) are different across the `RiskLevel` groups. This tells us that there are separable differences between the groups, making a classification model like LDA or QDA a reasonable pursuit.

In part (B), the significant Box's M test result indicated that the covariance (or, the dispersion) matrices are unequal across the three `RiskLevel` groups. This directly violates a primary statistical assumption of LDA, which requires that all groups have equal (homogeneous) covariance matrices.

QDA is specifically designed for this exact scenario. It does not assume equal covariance matrices and instead fits a more flexible quadratic boundary by modeling each group's dispersion separately. Using LDA when its assumptions are clearly violated would be statistically inappropriate and would likely lead to a less accurate classifier.

It is important to note that while our tests on the means (MANOVA) and dispersions (Box's M) clearly point us to QDA over LDA, we technically have not yet formally checked the other key assumption shared by both models: that the predictors are multivariate normal within each `RiskLevel` group.

Question 3: Discriminant Analysis (LDA/QDA) (15 Points)

Based on your findings in question 2, please build the appropriate discriminant analysis model.

Part A (2 points) Split the data into a training set (80%) and a test set (20%).

```r
set.seed(42)
trainIndex <- createDataPartition(data$RiskLevel, p = 0.8, list = FALSE,
    times = 1)
train_set <- data[trainIndex, ]
train_x <- as.matrix(train_set |>
    dplyr::select(-RiskLevel))
train_y <- train_set |>
    dplyr::pull(RiskLevel)

test_set <- data[-trainIndex, ]
test_x <- as.matrix(test_set |>
    dplyr::select(-RiskLevel))
test_y <- test_set |>
    dplyr::pull(RiskLevel)
```

Part B (3 points) Train your chosen discriminant model on the training set and generate predictions on the test set.

```r
### So we must do QDA
qda_model <- MASS::qda(x = train_x, grouping = train_y)
qda_preds <- predict(qda_model, newdata = test_x)$class
```

Part C (3 points) Create and display the confusion matrix and all associated statistics by using `confusionMatrix()`

```r
### how did the model do
caret::confusionMatrix(reference = factor(test_y), data = qda_preds)
```

```
## Confusion Matrix and Statistics
##
##            Reference
## Prediction  high risk low risk mid risk
##   high risk        37        2        4
##   low risk          9       68       44
##   mid risk          8       11       19
##
## Overall Statistics
##
##                Accuracy : 0.6139
##                  95% CI : (0.543, 0.6814)
##     No Information Rate : 0.401
##     P-Value [Acc > NIR] : 8.561e-10
##
##                   Kappa : 0.3971
##
##  Mcnemar's Test P-Value : 1.163e-05
##
## Statistics by Class:
##
##                      Class: high risk Class: low risk Class: mid risk
## Sensitivity                    0.6852          0.8395         0.28358
## Specificity                    0.9595          0.5620         0.85926
## Pos Pred Value                 0.8605          0.5620         0.50000
## Neg Pred Value                 0.8931          0.8395         0.70732
## Prevalence                     0.2673          0.4010         0.33168
## Detection Rate                 0.1832          0.3366         0.09406
## Detection Prevalence           0.2129          0.5990         0.18812
## Balanced Accuracy              0.8223          0.7007         0.57142
```

Part D (2 points) What is the model's `Overall Accuracy` on the test set? (2 points)

Overall, the model correctly classified 61.4% (95% CI, 54.3%-68.1%) of the patients in our test set. It's significantly better than the `No Information Rate` of 40.1% (which is just guessing the most common class, `low risk`, every time).

Part E (5 points) Look at the `Statistics by Class`. Comment on the model's strengths and weaknesses?

- **Strengths**
  - **High Specificity for `high risk` (95.95%): The model is very strong at correctly identifying patients who are not `high risk`. When a patient is truly `low` or `mid` risk, the**

model correctly avoids labeling them as `high risk` 95.95% of the time.
  - **High Positive Predictive Value for `high risk` (86.05%): When the model does predict a patient is `high risk`, it is correct 86.05% of the time. This makes its `high risk` predictions quite reliable.**
  - **Good Sensitivity for `low risk` (83.95%): The model is reasonably successful at finding the `low risk` patients, correctly identifying 83.95% of all actual `low risk` individuals.**

- **Weaknesses**
  - **Extremely Poor Sensitivity for `mid risk` (28.35%): This is the model's biggest failure. It only finds 28.35% of the actual `mid risk` patients. This means the remaining 71.65% of all `mid risk` patients are being misclassified as either `high` or `low` risk.**
  - **Poor Specificity for `low risk` (56.20%): This is the other side of the `mid risk` problem. The model often misclassifies `mid` and `high` risk patients as `low risk`. This is a significant issue, as it could lead to under-treating patients who need serious treatment.**
  - **Unreliable `mid risk` Predictions (Pos Pred Value: 50.00%): When the model predicts a patient is `mid risk`, it is only correct 50% of the time. This prediction is no better than a coin toss.**

Question 4: K-Nearest Neighbors (k-NN) (15 Points)

Let's see if a non-parametric model can perform better.

Part A (7 points) Using the same train-test splits you created in question 3, train a k-nn model. **You must scale your data (e.g., using the `preProcess = c("center", "scale")` argument within the train function).** Use 10-fold cross-validation to automatically tune the `k` parameter.

```r
# Okay so QDA isn't that nice, lets do k-nn instead...
set.seed(42)
ctrl <- trainControl(method = "cv", number = 10) # 10-fold cross-validation
knn_model <- train(
  x = train_x, y = train_y,
  method = "knn",
  trControl = ctrl,
  preProcess = c("center", "scale"), # Scaling is important for k-NN
  tuneGrid = expand.grid(k = 1:42)) ## try many values of k
knn_preds <- predict(knn_model, newdata = test_x)
```

Part B (3 points) Generate and display the confusion matrix from the model's performance on the test set.

```r
### how does k-nn do?
caret::confusionMatrix(reference = factor(test_y), data = knn_preds)
```

```
## Confusion Matrix and Statistics
##
##            Reference
## Prediction  high risk low risk mid risk
##   high risk        47        2        4
##   low risk          4       63       19
##   mid risk          3       16       44
##
## Overall Statistics
##
##                Accuracy : 0.7624
##                  95% CI : (0.6976, 0.8193)
```

```
##     No Information Rate : 0.401
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.6376
##
##  Mcnemar's Test P-Value : 0.7851
##
## Statistics by Class:
##
##                      Class: high risk Class: low risk Class: mid risk
## Sensitivity                    0.8704          0.7778          0.6567
## Specificity                    0.9595          0.8099          0.8593
## Pos Pred Value                 0.8868          0.7326          0.6984
## Neg Pred Value                 0.9530          0.8448          0.8345
## Prevalence                     0.2673          0.4010          0.3317
## Detection Rate                 0.2327          0.3119          0.2178
## Detection Prevalence           0.2624          0.4257          0.3119
## Balanced Accuracy              0.9149          0.7938          0.7580
```

Part C (1 + 1 + 3 = 5 points)

(i) What is the overall accuracy of the optimized k-NN model?
(ii) How does it compare to the discriminant model from question 3?
(iii) Look at the specific errors in the confusion matrix. From a clinical safety perspective, what is the single most dangerous type of error this model is making, and how many times did it make that error in the test set?

Overall, the model correctly classified 76.2% (95% CI, 69.8%-82.0%) of the patients in our test set. The k-NN model's accuracy of 76.2% is higher than the 61.4% accuracy achieved by the QDA model. From a clinical safety perspective, the single most dangerous type of error is misclassifying a patient who is truly `high risk` as being `mid risk` or `low risk`. This error could lead to a critical lack of necessary treatment. Looking at the confusion matrix, the model made this specific error 7 times in the test set.

Question 5: Unsupervised Clustering (20 Points)

In this section, we will ignore the `RiskLevel` labels and use unsupervised methods to see if the data naturally forms groups. Using only the 6 predictor variables (no `RiskLevel`) we will build an unsupervised clustering model. **Scale this data. This is critical for clustering.** For this question, use the whole dataset instead of training and testing splits.

Part A (6 + 6 + 3 = 15 points)

(i) Perform K-Means clustering with `centers = 3`.
(ii) Perform Hierarchical clustering (use `average` dissimilarity on a `Euclidean` distance matrix) and cut the resulting tree into `k = 3` clusters.
(iii) For both clustering results, create a cross-tabulation (a `table()`) that compares the true RiskLevel labels to the cluster assignments.

```r
x <- as.matrix(data |>
    dplyr::select(-"RiskLevel"))
y <- data |>
    dplyr::pull("RiskLevel")
### k-means
set.seed(42)
```

```r
km_model <- kmeans(scale(x), centers = 3, algorithm = "Lloyd")
### hierarchical
dist_matrix <- dist(scale(x), method = "euclidean")  ## eucli
hclust_model <- hclust(dist_matrix, method = "average")  ## average dissimilarity
hclust_clusters <- cutree(hclust_model, k = 3)  ## cut into three clusters
```

Part B (5 points) Based on the tables from part (a) above, did either clustering algorithm successfully recover the true `RiskLevel` categories? What does this tell you about the structure of the data and the separability of each of the risk groups?

```r
table(True = y, K_Means = km_model$cluster)
```

```
##            K_Means
## True          1   2   3
##   high risk  63  61 148
##   low risk  245 156   5
##   mid risk  198 106  32
```

```r
table(True = y, Hclust = hclust_clusters)
```

```
##            Hclust
## True          1   2   3
##   high risk 111 161   0
##   low risk    0 404   2
##   mid risk   24 312   0
```

No, neither clustering algorithm fully recovered all three true `RiskLevel` categories. Both algorithms showed partial success but ultimately failed to separate all groups cleanly.

The `high risk` group is at least partially separable from the other two groups. Its data structure appears to be distinct enough that both unsupervised algorithms could identify a cluster predominantly containing these patients. The `low risk` and `mid risk` groups are not easily separable using clustering. They likely share a very similar structure in their features (in the current), causing the (unsupervised) clustering algorithms to fail to find a natural boundary between them and grouping them together. We could consider transforming features to either functions of the existing features or to higher dimensional spaces.